



Gibson Assembly

Designing Life with AI

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Description

This method is used for assembling new plasmids by joining multiple DNA fragments together. It is performed following 3 main steps:

- **I – PCR amplification** to linearize the vectors and generate the overhangs needed for insertion.
- **II – PCR clean up** to get rid of all the PCR residues.
- **III – Cloning Assembly** to assemble the different fragments together and obtain a circular plasmid.

Steps

I – Primers design for cloning

Before starting any experiment, primers with overhangs need to be designed as they will allow the assembly of the different fragments during the cloning step. Here is a guideline of things to consider to make the most efficient primers possible (we designed them on Benchling):

- **Directionality in Benchling:**
 - Top strand is $5' \rightarrow 3'$ (starts with ATG); bottom strand is $3' \rightarrow 5'$.
 - Forward primer = binds top strand, faces right.
 - Reverse primer = binds bottom strand, faces left.
 - Tip: to flip plasmid orientation, select entire sequence $\rightarrow$ copy $\rightarrow$ paste reverse complement $\rightarrow$ Ctrl+V (keep annotations!).
- **BE CAREFUL WITH THE DIRECTIONALITY**, knowing that PCR always goes from $5'$ to $3'$.
- **Start/stop codons:**
 - Start codon = ATG (Met).
 - Stop codons = UGA, UAA, UAG (appear as * in Benchling).
- **Primer length:**
 - 18–30 bp (without overhangs).
 - Longer primers reduce chance of off-target binding.
- **Melting temperature (T_m):**
 - Do not go below 52°C (too unspecific).
 - High GC raises T_m ; 72°C is workable but demanding.
 - Aim for $\sim 60^\circ\text{C}$ T_m with $\sim 54^\circ\text{C}$ annealing.
 - T_m is more important than length.
- **GC content:**
 - Ideal between 45–60%.
 - Try to end primers with G or C (“GC clamp”).
- **Secondary structures:**
 - Avoid homodimers (primers binding each other).
 - Avoid hairpins/monomers (self-binding).

- $\Delta G < -13$ kcal/mol = too stable, likely to form → redesign.
- **Overhangs for cloning:**
 - Add ~30 bp overhangs at the 5' end of primers (never at 3').
 - Use neutral sequences (no promoter, terminator, RBS).
 - Inform Benchling about added bp to avoid mismatch alerts.
 - Check in Benchling that overhangs are present and appear exactly twice, side-by-side, and on the same strand.
 - Neutral sequences: use “spacers” from Schaerli plasmid design or consult team.
 - Alternative: add a homology sequence instead of extra overhangs.
- **Fragment size:**
 - Best to amplify ≤ 5 kb.
 - For longer fragments, split into 2 pieces and amplify separately.
- **Plasmid assembly verification in Benchling:**
 - Export the PCR product sequence.
 - Select the full PCR product in Benchling.
 - Do not include the overhangs (OH) when copying.
 - If the OH appear in bold, Benchling interprets them as part of the annotation — this means they're already “taken.” Ignore them for now.
- **Create a new DNA file:**
 - In Benchling, go to DNA → New DNA.
 - Paste only the primer binding regions (without OH) into this new sequence.
 - This file will serve as a “clean” version to test assembly.
- **Check overhangs manually:**
 - Copy each overhang sequence separately.
 - Paste them into the new DNA file to search for matches.
 - The OH should appear exactly 2 times in the file (not more, not less), side by side (directly adjacent), and on the same strand.
 - If any of these conditions fail, the assembly will not work properly.

II – PCR amplification (to create linear fragments and amplify them)

We used the Phanta Max Master Mix, following the manufacturer's instructions:

<https://www.vazymeglobal.com/product-center/high-fidelity-pcr/2-phanta-max-master-mix-dye-plus>

1. Put the following recipe in PCR tubes:

Ingredients	Volume [μl]
2x Taq or Q5 HF Master Mix	25
Forward primer	2.5
Reverse primer	2.5
DNA template	1
MilliQ water	19
Total	50

2. Use the PCR program from manufacturer and modify annealing temperature and elongation time according to the primers used.

III – Fragment length verification

1. Prepare agarose gel:
 - (a) Weigh 0.5 gr of agarose (to get to 1%)
 - (b) Mix with 50 ml of 1x TBE buffer
 - (c) Heat it 1 minute and 30 seconds in the microwave
 - (d) Let it cool a bit
 - (e) Add 1 µl of Sybrsafe (1 µl per 50 ml gel) and mix gently
 - (f) Pour the gel in a rack with the appropriate comb
 - (g) Let it solidify for around 20 min
2. Load 5 µl of DNA ladder and PCR product + add 1 µl of loading dye per sample if necessary.
3. Run gel at 135 V for 15 min.
4. Visualize DNA bands under UV light.

IV – PCR clean up

Clean the remaining 45 µl of the PCR product using the Monarch PCR & DNA Clean-up Kit following the manufacturer's instructions:

<https://www.neb.com/en/products/t1030-monarch-pcr-dna-cleanup-kit-5-ug>

Protocol: All centrifugation steps should be carried out at 13'000 rpm.

1. Dilute sample with DNA Cleanup Binding Buffer according to the table below. Mix well by pipetting up and down or flicking the tube. Do not vortex.
2. Insert column into collection tube and load sample onto column. Spin for 1 minute, then discard flow-through.
3. Add 200 µl DNA Wash Buffer and spin for 1 minute.
4. Repeat wash.
5. Transfer column to a clean 1.5 ml microfuge tube.
6. Add 15 µl of ddH₂O. Wait for 1 minute, then spin for 1 minute to elute DNA.
7. Perform a DNA concentration and purity measurement with a nanodrop machine. If DNA purity is too low, perform the washing steps again.

Sample type	Ratio of binding buffer : sample	Example
dsDNA > 2 kb (plasmids, gDNA)	2:1	200 µl : 100 µl
dsDNA < 2 kb (some amplicons, fragments)	5:1	500 µl : 100 µl
ssDNA > 200 nt	7:1	700 µl : 100 µl

IV – Optimal concentrations calculation

1. Measure the DNA concentration using a nanodrop machine.
2. Based on the following formula:

$$(\text{insert mass}) = 100\text{ng} \times (\text{molarity multiplier factor}) \times (\text{insert size}) / (\text{backbone size})$$

Molarity multiplier factor: 2 if (insert size) > 200 bp; 5 if (insert size) < 200 bp.

Calculate the optimal inserts concentration, knowing that we want 100 ng/μl of backbone (the longest fragment). Using the provided Excel template, DNA concentration from nanodrop and fragment length just need to be inserted to obtain the volumes to add to reach the optimal concentration.

V – Gibson cloning

1. Prepare the solutions with the optimal concentrations of the different fragments.
2. In PCR tubes, prepare the following recipe based on the NEBuilder HiFi DNA Master Mix manufacturer's instructions:

<https://www.neb.com/en/products/e2621-nebuilder-hifi-dna-assembly-master-mix>

Ingredients	Volumes [μl]
NEBuilder HiFi DNA Master Mix	10
Backbone	1
Insert(s)	1 per insert
ddH ₂ O	up to 20
Total	20

3. In parallel, prepare a control reaction containing the same recipe but with 1 insert less.
4. Incubate both tubes at 50 °C for 1 hour.

The Gibson assembly is finally done. The following steps explain chronologically what to do with the newly made plasmid.

VI – Transformation

The next step is to introduce the fresh plasmid into competent cells (see competent cells protocol) of the *E. coli* DH5α cloning strain (see transformation protocol). Note that once the plasmid is verified, it can be transformed into the “working strain”, in our case it was the *E. coli* BL21.

VII – Colony PCR

Perform a Colony PCR to verify if the plasmid has been correctly inserted (link to colony PCR protocol).

VIII – Sequencing

In case of positive control from the Colony PCR, send the PCR product to be sequenced to verify if the DNA sequence of the plasmid is correct (see preparation for sequencing protocol).

IX – Liquid culture

Make an overnight liquid culture of the colonies sent for sequencing; in case of positive result, glycerol stock and plasmid extraction will be performed on it (see liquid culture preparation protocol).

X – Stocks

- Make a glycerol stock of the transformed bacteria for further experiments (see glycerol stock protocol).
- Perform a miniprep to extract the plasmid in order to insert it in another strain (see miniprep protocol).